

1. USV and controller parameters

First, given the matrices A_η , A_v and B_{uw} as:

$$\begin{aligned} A_\eta &= -M^{-1}G = \begin{bmatrix} -0.0358 & 0 & 0 \\ 0 & -0.0208 & 0 \\ 0 & -0.0394 & 0 \end{bmatrix}, \\ A_v &= -M^{-1}N = \begin{bmatrix} -0.0797 & 0 & 0 \\ 0 & -0.0818 & -0.0577 \\ 0 & -0.2254 & -0.2535 \end{bmatrix}, \\ B_{uw} &= M^{-1} = \begin{bmatrix} 0.9215 & 0 & 0 \\ 0 & 0.7802 & 1.4811 \\ 0 & 1.4811 & 7.4562 \end{bmatrix}. \end{aligned}$$

Accordingly, for $\theta(t) \in [-\frac{\pi}{6}, \frac{\pi}{6}]$, the nonlinear USV can be modeled by the subsequent local T-S fuzzy systems:

$$\begin{aligned} A_i &= \begin{bmatrix} 0 & J_i \\ A_\eta & A_v \end{bmatrix}, \quad B_i = \begin{bmatrix} 0 \\ B_{uw} \end{bmatrix}, \\ C_{2i} &= [0 \quad 0 \quad -0.002 \quad 0 \quad 0 \quad 0.01], i \in \{1, 2, 3, 4\} \end{aligned}$$

where the corresponding rotational matrices are defined as:

$$\begin{aligned} J_1 &= \begin{bmatrix} 1 & -1/2 & 0 \\ 1/2 & -1/2 & 0 \\ 0 & 1 & 1 \end{bmatrix}, J_2 = \begin{bmatrix} \sqrt{3}/2 & -1/2 & 0 \\ 1/2 & \sqrt{3}/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \\ J_3 &= \begin{bmatrix} 1 & 1/2 & 0 \\ -1/2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, J_4 = \begin{bmatrix} \sqrt{3}/2 & 1/2 & 0 \\ -1/2 & \sqrt{3}/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \end{aligned}$$

Furthermore, the FMF are explicitly constructed as:

$$\begin{aligned} g_1(\theta(t)) &= \phi_1(x_3(t))(-3 - 2\sqrt{3} + (4 + 2\sqrt{3})\cos(x_3(t))), \\ g_2(\theta(t)) &= \phi_1(x_3(t))(4 + 2\sqrt{3} - (4 + 2\sqrt{3})\cos(x_3(t))), \\ g_3(\theta(t)) &= \phi_2(x_3(t))(-3 - 2\sqrt{3} + (4 + 2\sqrt{3})\cos(x_3(t))), \\ g_4(\theta(t)) &= \phi_2(x_3(t))(4 + 2\sqrt{3} - (4 + 2\sqrt{3})\cos(x_3(t))), \end{aligned}$$

in which the intermediate variables ϕ_1 and ϕ_2 are given by:

$$\phi_1(x_3(t)) = \frac{1}{2} + \sin(x_3(t)), \quad \phi_2(x_3(t)) = \frac{1}{2} - \sin(x_3(t)).$$

Meanwhile, the time-varying environmental disturbances $w(t)$ are described as:

$$\begin{cases} w_1(t) = -0.1e^{-t} \\ w_2(t) = -0.4\cos(1.58t)e^{-0.235t} \\ w_3(t) = -0.2\sin(2t)e^{-2t} \end{cases} \quad (1)$$

The feedback control gains alongside the triggering weight matrices are computed as follows:

$$\begin{aligned} K_{11} &= \begin{bmatrix} -1.1990 & -0.1299 & -0.0041 & -1.7782 & -0.0696 & -0.0069 \\ 0.2418 & -2.2834 & 0.3883 & 0.1289 & -3.4120 & 0.5784 \\ -0.0469 & 0.4520 & -0.1862 & -0.0252 & 0.6895 & -0.2595 \end{bmatrix}, \\ K_{12} &= \begin{bmatrix} -0.7055 & -0.0745 & -0.0024 & -1.0478 & -0.0397 & -0.0041 \\ 0.1387 & -1.3435 & 0.2293 & 0.0735 & -2.0107 & 0.3418 \\ -0.0269 & 0.2660 & -0.1097 & -0.0144 & 0.4064 & -0.1530 \end{bmatrix}, \end{aligned}$$

$$K_{13} = \begin{bmatrix} -0.3659 & -0.0373 & -0.0013 & -0.5444 & -0.0197 & -0.0022 \\ 0.0694 & -0.6970 & 0.1193 & 0.0364 & -1.0447 & 0.1780 \\ -0.0135 & 0.1380 & -0.0569 & -0.0071 & 0.2112 & -0.0796 \end{bmatrix},$$

$$K_{21} = \begin{bmatrix} -1.2142 & -0.1581 & -0.0065 & -1.8267 & -0.0845 & -0.0100 \\ 0.2942 & -2.3114 & 0.3894 & 0.1573 & -3.5013 & 0.5824 \\ -0.0573 & 0.4575 & -0.1865 & -0.0312 & 0.7070 & -0.2603 \end{bmatrix},$$

$$K_{22} = \begin{bmatrix} -0.7143 & -0.0905 & -0.0039 & -1.0759 & -0.0481 & -0.0059 \\ 0.1683 & -1.3599 & 0.2298 & 0.0895 & -2.0624 & 0.3440 \\ -0.0328 & 0.2693 & -0.1098 & -0.0177 & 0.4166 & -0.1535 \end{bmatrix},$$

$$K_{23} = \begin{bmatrix} -0.3706 & -0.0458 & -0.0021 & -0.5590 & -0.0242 & -0.0031 \\ 0.0852 & -0.7056 & 0.1194 & 0.0450 & -1.0716 & 0.1790 \\ -0.0166 & 0.1398 & -0.0570 & -0.0089 & 0.2165 & -0.0798 \end{bmatrix},$$

$$K_{31} = \begin{bmatrix} -1.1990 & 0.1299 & 0.0041 & -1.7782 & 0.0696 & 0.0069 \\ -0.2418 & -2.2834 & 0.3883 & -0.1289 & -3.4120 & 0.5784 \\ 0.0469 & 0.4520 & -0.1862 & 0.0252 & 0.6895 & -0.2595 \end{bmatrix},$$

$$K_{32} = \begin{bmatrix} -0.7055 & 0.0745 & 0.0024 & -1.0478 & 0.0397 & 0.0041 \\ -0.1387 & -1.3435 & 0.2293 & -0.0735 & -2.0107 & 0.3418 \\ 0.0269 & 0.2660 & -0.1097 & 0.0144 & 0.4064 & -0.1530 \end{bmatrix},$$

$$K_{33} = \begin{bmatrix} -0.3659 & 0.0373 & 0.0013 & -0.5444 & 0.0197 & 0.0022 \\ -0.0694 & -0.6970 & 0.1193 & -0.0364 & -1.0447 & 0.1780 \\ 0.0135 & 0.1380 & -0.0569 & 0.0071 & 0.2112 & -0.0796 \end{bmatrix},$$

$$K_{41} = \begin{bmatrix} -1.2142 & 0.1581 & 0.0065 & -1.8267 & 0.0845 & 0.0100 \\ -0.2942 & -2.3114 & 0.3894 & -0.1573 & -3.5013 & 0.5824 \\ 0.0573 & 0.4575 & -0.1865 & 0.0312 & 0.7070 & -0.2603 \end{bmatrix},$$

$$K_{42} = \begin{bmatrix} -0.7143 & 0.0905 & 0.0039 & -1.0759 & 0.0481 & 0.0059 \\ -0.1683 & -1.3599 & 0.2298 & -0.0895 & -2.0624 & 0.3440 \\ 0.0328 & 0.2693 & -0.1098 & 0.0177 & 0.4166 & -0.1535 \end{bmatrix},$$

$$K_{43} = \begin{bmatrix} -0.3706 & 0.0458 & 0.0021 & -0.5590 & 0.0242 & 0.0031 \\ -0.0852 & -0.7056 & 0.1194 & -0.0450 & -1.0716 & 0.1790 \\ 0.0166 & 0.1398 & -0.0570 & 0.0089 & 0.2165 & -0.0798 \end{bmatrix}.$$

$$\Psi_1 = \begin{bmatrix} 0.3808 & -0.0003 & -0.0009 & 0.5368 & -0.0004 & -0.0005 \\ -0.0003 & 0.3846 & -0.0067 & 0.0001 & 0.5400 & -0.0173 \\ -0.0009 & -0.0067 & 0.1932 & -0.0002 & -0.0319 & 0.2455 \\ 0.5368 & 0.0001 & -0.0002 & 0.8082 & -0.0000 & 0.0001 \\ -0.0004 & 0.5400 & -0.0319 & -0.0000 & 0.8120 & -0.0553 \\ -0.0005 & -0.0173 & 0.2455 & 0.0001 & -0.0553 & 0.3342 \end{bmatrix},$$

$$\Psi_2 = \begin{bmatrix} 0.3812 & -0.0004 & -0.0006 & 0.5543 & -0.0019 & -0.0000 \\ -0.0004 & 0.3847 & -0.0071 & 0.0016 & 0.5570 & -0.0184 \\ -0.0006 & -0.0071 & 0.1932 & 0.0005 & -0.0320 & 0.2456 \\ 0.5543 & 0.0016 & 0.0005 & 0.8429 & -0.0001 & 0.0010 \\ -0.0019 & 0.5570 & -0.0320 & -0.0001 & 0.8459 & -0.0562 \\ -0.0000 & -0.0184 & 0.2456 & 0.0010 & -0.0562 & 0.3343 \end{bmatrix},$$

$$\Psi_3 = \begin{bmatrix} 0.3808 & 0.0003 & 0.0009 & 0.5368 & 0.0004 & 0.0005 \\ 0.0003 & 0.3846 & -0.0067 & -0.0001 & 0.5400 & -0.0173 \\ 0.0009 & -0.0067 & 0.1932 & 0.0002 & -0.0319 & 0.2455 \\ 0.5368 & -0.0001 & 0.0002 & 0.8082 & 0.0000 & -0.0001 \\ 0.0004 & 0.5400 & -0.0319 & 0.0000 & 0.8120 & -0.0553 \\ 0.0005 & -0.0173 & 0.2455 & -0.0001 & -0.0553 & 0.3342 \end{bmatrix},$$

$$\Psi_4 = \begin{bmatrix} 0.3812 & 0.0004 & 0.0006 & 0.5543 & 0.0019 & 0.0000 \\ 0.0004 & 0.3847 & -0.0071 & -0.0016 & 0.5570 & -0.0184 \\ 0.0006 & -0.0071 & 0.1932 & -0.0005 & -0.0320 & 0.2456 \\ 0.5543 & -0.0016 & -0.0005 & 0.8429 & 0.0001 & -0.0010 \\ 0.0019 & 0.5570 & -0.0320 & 0.0001 & 0.8459 & -0.0562 \\ 0.0000 & -0.0184 & 0.2456 & -0.0010 & -0.0562 & 0.3343 \end{bmatrix}.$$